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Algebra Lineal

Grupo 17

Semestre 2022-2

Tarea 2

Sea la matriz

$$M(T) = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

la representación matricial del operador lineal $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Determinar los valores y vectores característicos asociados al operador y, si es posible, verificar

$D = P^{-1} M(T) P$

10.- D

1.- $|M(T) - \lambda I| = 0$

2.- Raíces

3.- Ecuación característica

4.- V.C.

5.- B

6.- UB

7.- P

8.- P^{-1}

9.- $P^{-1} M(T) \cdot P$

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$$|M(T) - \lambda I| = 0$$

$$M(T) = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} \qquad I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\lambda I = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & \lambda & 0 \end{pmatrix} = \begin{vmatrix} -1-\lambda & 2 & 0 \\ 0 & 3-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix}$$

$$\left|\begin{matrix} -1-\lambda & 2 & 0 & -1-\lambda & 2 \\ 0 & 3-\lambda & 1 & 0 & 3-\lambda \\ 0 & 0 & 4-\lambda & 0 & 0 \end{matrix}\right|$$

$$= (-1-\lambda)(3-\lambda)(4-\lambda) + (0) + (0) + (0) + (0) + (0)$$

$$= (-3 + \lambda - 3\lambda + \lambda^2)(4-\lambda)$$

$$= -12 + 4\lambda - 12\lambda + 4\lambda^2 + 3\lambda - \lambda^2 + 3\lambda^2 - \lambda^3$$

$$= -5\lambda + 6\lambda^2 - \lambda^3 - 12$$

$$-\lambda^3 + 6\lambda^2 - 5\lambda - 12 = 0 \rightarrow \text{Polinomio característico de } T$$

$$-\lambda^3 + 6\lambda^2 - 5\lambda - 12 = 0$$

$$(-\lambda - 1)(\lambda - 3)(\lambda - 4) = 0$$

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$-\lambda - 1 = 0$
$-\lambda = 1$
$\underline{\lambda_1 = -1}$ //

$\lambda - 3 = 0$
$\underline{\lambda_2 = 3}$ //

$\lambda - 4 = 0$
$\underline{\lambda_3 = 4}$ // $\rightarrow$ Valores característicos de T

$$(M(T) - \lambda I)[\bar{v}]_B = 0$$

Para $\lambda_1 = -1$

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} - (-1)\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

$[\bar{v}]_B$

$[(x, y, z)]_B$ $\qquad B = \{(1,0,0), (0,1,0), (0,0,1)\}$

$(x, y, z) = \alpha_1(1,0,0) + \alpha_2(0,1,0) + \alpha_3(0,0,1)$

$(x, y, z) = (\alpha_1, \alpha_2, \alpha_3)$

$$[\bar{v}]_B = [(x,y,z)]_B = \underline{\begin{pmatrix} x \\ y \\ z \end{pmatrix}} //$$

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$2y = 0$
$\underline{y = 0}$ //

$4y + z = 0$
$4(0) + z = 0$
$\underline{z = 0}$ //

$5z = 0$
$\underline{z = 0}$ //

$\underline{x = x}$ //

$V.C. = \{(x, y, z) \mid x, y, z \in R\}$

$V.C(\lambda_1 = -1) = \{(x, 0, 0) \mid x \in R\}$ // Ecuación característica λ_1

Para $\lambda_2 = 3$

$(M(T) - \lambda I)[\bar{V}]_B$

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -4 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$-4x + 2y = 0 \longrightarrow 2y = 4x$

$z = 0$ // $y = 2x$ //

$z = 0$ //

$-4x + 2(2x) = 0$

$\cancel{-4x} + \cancel{4x} = 0$

$x = x$ //

$V.C. = \{(x, y, z) \mid x, y, z \in IR\}$

$V.C. = (\lambda_2 = 3) = \{(x, 2x, 0) \mid x \in IR\}$ //

Ecuación característica λ_2

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Para $\lambda_3 = 4$

$(M(T) - \lambda I)[\bar{v}]_B$

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} - 4\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} -5 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}} = 0$$

$-5x + 2y = 0 \longrightarrow 2y = 5x$

$y = \frac{5}{2}x$

$-y + z = 0$

$-\left(\frac{5}{2}x\right) + z = 0$

$z = \frac{5}{2}x$

$-5x + 2\left(\frac{5}{2}x\right) = 0$

$-5x + 5x = 0$

$x = x$

$V.C. = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$

$V.C. = (\lambda_3 = 4) = \{(x, \frac{5}{2}x, \frac{5}{2}x) \mid x \in \mathbb{R}\}$

Ecuación característica λ_3

$V.C.\ \lambda_1 = -1 = \{(x, 0, 0) \mid x \in \mathbb{R}\}$

$B_{\lambda_1} = \{(1, 0, 0)\}$

V.C. $\lambda_2 = 3 = \{(x, 2x, 0) \mid x \in \mathbb{R}\}$

$B_{\lambda_2} = \{(1, 2, 0)\}$

V.C. $\lambda_3 = 4 = \{(x, \frac{5}{2}x, \frac{5}{2}x) \mid x \in \mathbb{R}\}$

$B_{\lambda_3} = \{(1, \frac{5}{2}, \frac{5}{2})\}$

$UB = \{(1,0,0)^T, (1,2,0)^T, (1, \frac{5}{2}, \frac{5}{2})^T\}$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 5/2 \\ 0 & 0 & 5/2 \end{pmatrix} \to \text{Matriz diagonalizadora}$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 5/2 & 0 & 1 & 0 \\ 0 & 0 & 5/2 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} C_2 <=> C_1 \\ C_2 <=> C_3 \end{array} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 5/2 & 0 & 0 & 1 & 0 \\ 0 & 5/2 & 0 & 0 & 0 & 2/5 \end{array}\right)$$

$R_2 <=> R_3 (\frac{2}{5})$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 2/5 \\ 2 & 5/2 & 0 & 0 & 1 & 0 \end{array}\right) \begin{array}{l} R_2(-5/2) + R_3 = R_3 \\ R_2(-1) + R_1 = R_1 \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & -2/5 \\ 0 & 1 & 0 & 0 & 0 & 2/5 \\ 2 & 0 & 0 & 0 & 1 & -1 \end{array}\right)$$

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$R_3(1/2) + R_3(-1) + R_1 = R_1$

$$\left(\begin{array}{ccc|ccc} 0 & 0 & 1 & 1 & -1/2 & 1/10 \\ 0 & 1 & 0 & 0 & 0 & 2/5 \\ 1 & 0 & 0 & 0 & 1/2 & -1/2 \end{array}\right) R_3 \leftrightarrow R_1 \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1/2 & -1/2 \\ 0 & 1 & 0 & 0 & 0 & 2/5 \\ 0 & 0 & 1 & 1 & -1/2 & 1/10 \end{array}\right)$$

$$P^{-1} = \begin{pmatrix} 1 & -1/2 & \frac{1}{10} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 2/5 \end{pmatrix}$$

$D = P^{-1} M(T) P$

$$D = \begin{pmatrix} 1 & -1/2 & \frac{1}{10} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 2/5 \end{pmatrix} \begin{pmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 5/2 \\ 0 & 0 & 5/2 \end{pmatrix}$$

$$D = \begin{pmatrix} -1 & 1/2 & -1/10 \\ 0 & 3/2 & -3/2 \\ 0 & 0 & 8/5 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 5/2 \\ 0 & 0 & 5/2 \end{pmatrix}$$

$$D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

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